

# Project Evaluation Rubric

## Industrial Informatics

### 1. Project Objective

The final project aims to evaluate the competencies acquired during the **Industrial Informatics** course by developing a **software application using the C# programming language and the .NET ecosystem**.

Students must demonstrate the ability to:

- design and implement software applications using **Object-Oriented Programming (OOP)**;
- integrate applications with **databases**;
- apply **modern software architecture principles**;
- develop **structured, maintainable, and extensible code**.

### 2. General Requirements

The project must meet the following conditions:

- The **application logic (backend)** must be implemented in **C#**.
- The application must include a **user interface** (Windows Forms, WPF, ASP.NET, etc.).
- Use of **Object-Oriented Programming concepts**.
- Management of **persistent data** (database or XML).
- Clear **code organization** and **separation of responsibilities**.

### 3. Evaluation Rubric

| Level    | Application Complexity                       | Main Requirements                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade 7  | Basic Application                            | <ul style="list-style-type: none"> <li>• application developed in C# (.NET)</li> <li>• graphical interface using WinForms, WPF, WinUI, or ASP.NET</li> <li>• menu and at least 4 windows/pages</li> <li>• use of UI controls (textbox, combobox, checkbox, radiobutton, lists)</li> <li>• use of classes and objects (basic OOP)</li> </ul>                                                                        |
| Grade 8  | Application with Database                    | <ul style="list-style-type: none"> <li>• all requirements from Grade 7</li> <li>• integration of a database</li> <li>• at least 3 tables</li> <li>• implementation of CRUD operations (Create, Read, Update, Delete)</li> <li>• search and filtering functionalities</li> </ul>                                                                                                                                    |
| Grade 9  | Complex Application                          | <ul style="list-style-type: none"> <li>• all requirements from Grade 8</li> <li>• database management using Entity Framework or programmatic creation</li> <li>• relationships between tables</li> <li>• use of transactions for updating data</li> <li>• validation and business logic implemented in the backend</li> </ul>                                                                                      |
| Grade 10 | Advanced Application (Software Architecture) | <ul style="list-style-type: none"> <li>• all requirements from Grade 9</li> <li>• multitier architecture: <ul style="list-style-type: none"> <li>– presentation layer (UI)</li> <li>– business logic layer (C#)</li> <li>– data layer (DB or XML)</li> </ul> </li> <li>• application of SOLID principles</li> <li>• use of at least one Design Pattern</li> <li>• well-structured and maintainable code</li> </ul> |

## 4. Recommended Technologies

Students may use technologies from the **.NET ecosystem**, such as:

- **.NET**
  - **Windows Forms, MAUI, WPF**
  - **ASP.NET**
  - **Entity Framework**
  - **SQL Server / SQLite**
  - **XML**
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## 5. Example Project Topics

The application topic is chosen by the student. Possible examples include:

- **library management system**
  - **student management application**
  - **hotel reservation system**
  - **product management application**
  - **task management application**
  - **medical appointment scheduling system**
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## Notes

- The **complexity of the application** should reflect the topics studied during the course (**C#, OOP, databases, and software architecture**).
- Students are encouraged to apply **design principles and good programming practices**.